



1.1-cm-long thin-film lithium niobate Mach-Zehnder modulator with low driving voltage integrated by micro-transfer printing

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Abstract: The heterogeneous integration of thin-film lithium niobate (TFLN) provides energy-efficient broadband optical modulation of photonic platforms. Micro-transfer printing (μ -TP) yields dense integration of heterogeneous materials and cost-effective photonic integrated circuits. However, the reports so far have considered unpatterned slab membranes for the structure of the μ -TP integrated TFLN, and the short interaction length increases the driving voltage. In this study, the μ -TP of pre-structure-defined TFLN Mach-Zehnder (MZ) interferometer was demonstrated for up to 1.1 cm with propagation loss of 0.7 dB/cm. The TFLN MZ modulator integrated on a silicon substrate using μ -TP exhibited the lowest driving voltage V_{π} of 2.7 V and a modulation bandwidth of 28 GHz with a low insertion loss of 0.97 dB. In addition, modulation frequencies exceeding 50 GHz and V_{π} of 8.9 V were demonstrated in a device with 3-mm-long electrodes.

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1. Introduction

The rapid development of generative artificial intelligence technologies has been accelerating the demand for higher-bandwidth data center interconnects (DCI), thereby driving the transmission capacity of short-reach optical transceivers of up to 2 km from 800 G and 1.6 T toward 3.2 T [1]. Photonic integrated circuits (PICs), including silicon (Si) and silicon nitride (SiN) platforms, have become complex systems owing to mature complementary metal oxide semiconductor (CMOS) compatible processes. Electro-optic (EO) modulation is a key function of PICs, and Si photonics (SiPh) is an essential technology for realizing power-efficient and low-cost optical transceivers with high capacity and low latency. Si does not exhibit the Pockels effect because of its centrosymmetry, and thus free-carrier dispersion is employed for EO modulation [2]. However, the bandwidth of typical Si Mach-Zehnder modulators (MZMs) is approximately 40 GHz [3,4], making it challenging to reach a bandwidth beyond 100 GHz without increasing the optical absorption and degrading modulation efficiency [5]. This is because the p-n depletion Si modulator always suffers from the trade-off between an improvement in bandwidth and decrease in modulation efficiency, and the carrier mobility of Si is small compared with those of other semiconductors.

Heterogeneous integration of thin-film lithium niobate (TFLN) on SiPh is promising for scaling the bitrates of optical transceivers from the current 200 G/lane toward 400 G/lane [6]. This approach provides low-loss and energy-efficient EO modulation because TFLN exhibits no carrier absorption and a high EO coefficient [7,8]. The bandwidth of a TFLN modulator is restricted only by microwave losses of electrodes and substrates, reaching over 100 GHz under the condition of lightwave and microwave velocity matching. The material properties of LN are well known, allowing TFLN to be merged into the market with minimal investment in production

infrastructure. Notably, 800 G coherent transceivers based on TFLN modulators and SiPh receivers are already commercially available [9]. However, heterogeneous integration of TFLN into the SiPh platform still remains challenging. Heterogeneous Si/TFLN and SiN/TFLN hybrid modulators have been investigated using die-to-wafer or wafer-to-wafer bonding [10,11,12], which requires a large footprint of the bonded die or wafer. This makes the integration of multiple materials such as III-V compounds difficult although the chip area required for TFLN is small. In addition, the post-processing that includes removing the substrate and etching the LN, is CMOS-incompatible because of lithium contamination [13].

Micro-transfer printing (μ -TP) is a scalable and versatile method for heterogeneous integration [14]. Pre-fabricated coupons on the source wafer are selected and printed onto a local area of the target wafer using elastomer stamps, thereby enabling high-density flexible integration of multifunctional photonic devices, including III-V lasers [15] and magneto-optical isolators [16]. Here, the TFLN modulators can be integrated onto the SiPh chip surface after the back-end-of-line process of the CMOS foundries, which reduces the bonded area and avoids contamination. The μ -TP of unpatterned slab LN membrane [17,18] and pre-fabricated TFLN waveguide [19,20] have been reported. In the slab membrane, light is guided in the hybrid structure of the TFLN and Si or SiN, resulting in reduced confinement of optical power to the LN, weak EO interaction, and a large driving voltage. In the case of the pre-fabricated TFLN waveguide, only a microring resonator (MRR) with limited coupon size of $100 \times 100 \mu\text{m}^2$ has been reported. As no microelectrodes are formed in this area, a modulation function is not exhibited. Etched TFLN waveguides with strong optical LN confinement are promising solutions for reducing the driving voltage. They maximize the EO modulation efficiency and require a print of the long TFLN coupon. However, μ -TP of TFLN is a significant challenge with a device length of over 1 cm. This is attributed to the mechanical stability of the air-suspended coupons after the drying process.

In this study, we demonstrated μ -TP of the pre-waveguides-defined 1.1-cm-long TFLN MZMs onto a new Si target substrate. The design of monolithic LN tethers was optimized to stably suspend under-etched TFLN coupons, and the selected TFLN was bonded to a new Si target substrate by O_2 plasma-activated direct bonding. The strong optical confinement and long EO interaction length remarkably reduced the half-wave voltage to 2.7 V with an insertion loss of 0.97 dB.

2. Device fabrication

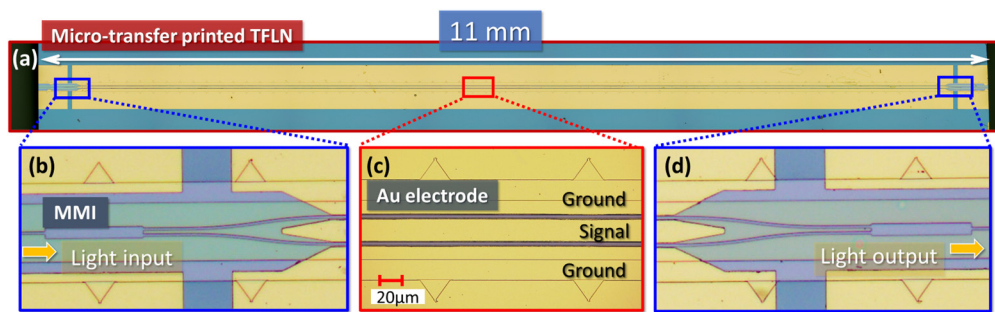


Fig. 1. Microscopic images of (a) overall view of micro-transfer printed TFLN MZM, (b) MMI coupler at input, (c) EO phase shifter having Au electrode with GSG configuration, and (d) MMI coupler at output.

A 1.1-cm-long TFLN MZM was fabricated on a Si substrate using μ -TP and a subsequent lift-off process. The microscopic image of the integrated TFLN MZM shown in Fig. 1 consists of two multi-mode interferometer (MMI) couplers at both the input and output ports (Figs. 1(b) and

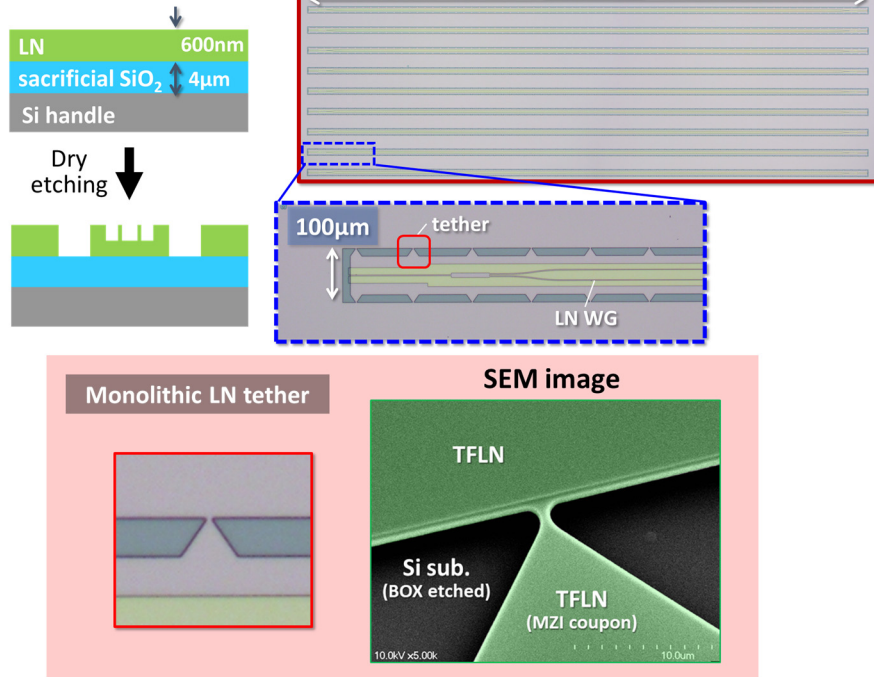
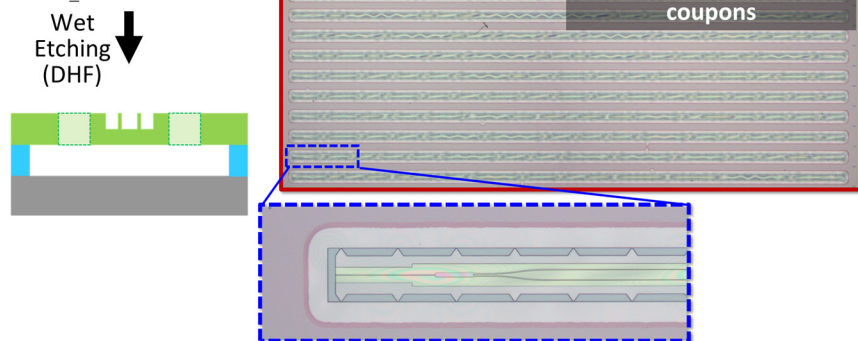
(a) Coupon fabrication**(b) SiO₂ removal**

Fig. 2. Fabrication process of free-standing 1.1-cm-long TFLN MZM coupons. (a) TFLN coupons fabricated by dry etching. The background of microscopic image and scanning electron microscopic (SEM) image of monolithic LN tethers are shaded in red. (b) Free-standing LN coupons underetched by DHF.

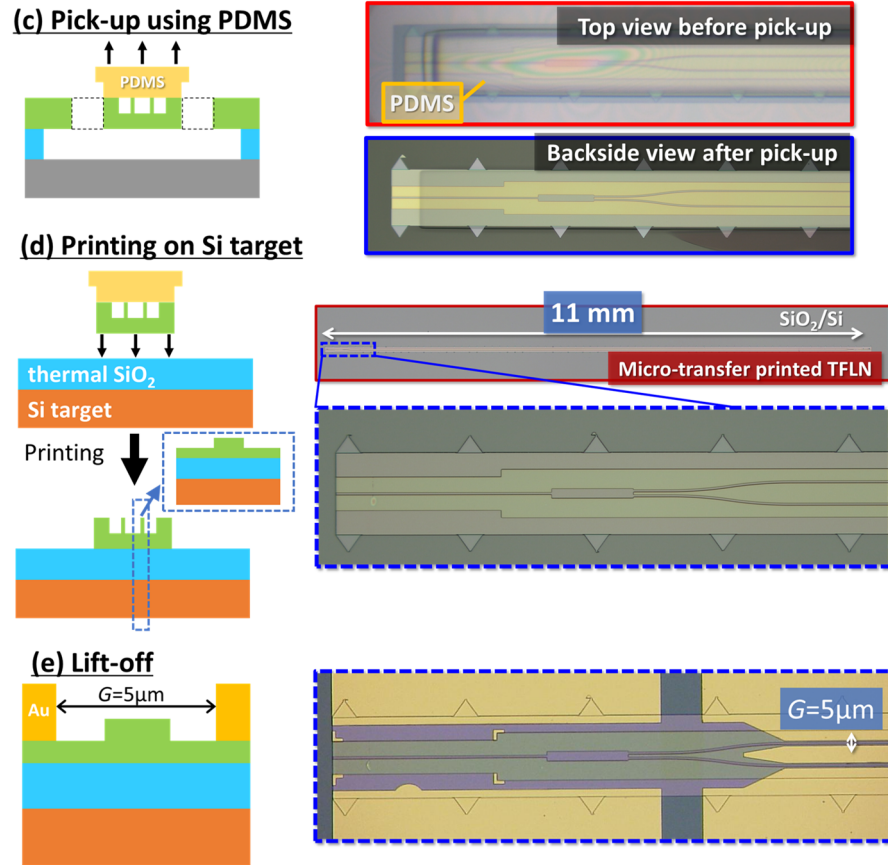


Fig. 3. μ -TP process of 1.1-cm-long TFLN MZM coupons. (c) pick-up process using PDMS stamp, (d) printing TFLM coupon on Si target substrate, and (e) lift-off process.

(d)) and arm waveguides (Fig. 1(c)), on which Au electrodes with ground-signal-ground (GSG) configuration are placed. The TFLN MZI was printed on a Si target substrate using the following process. TFLN MZIs were prefabricated on coupons before printing. First, a commercially available LN-on-insulator (LNOI) source wafer was prepared with a 600-nm-thick-X-cut TFLN layer, 4- μ m-thick sacrificial SiO_2 layer, and Si handle substrate. To maintain a stable suspension of TFLN coupons during printing, several support structure designs were investigated, including monolithic LN tethers [17,20], resist tethers [18,19,21], and pillar-based resist structures [22]. Monolithic LN tethers were employed for the coupons in this work because they neither require an additional resist layer nor a complex fabrication process for pillar structures. To suspend a large area of coupons, the design of LN tethers with sufficient mechanical strength must be optimized. Twelve channels of TFLN coupons with a footprint of $11 \times 0.1\text{-mm}^2$ were created on the LNOI chip, fabricating the MZIs using dry etching, as shown in Fig. 2(a). Around the MZIs, spaced monolithic tethers of 150 μm were formed in the TFLN layer using the optimized design for stable suspension, setting a minimum tether width of approximately 2.5 μm , to connect the coupons and source wafer. Subsequently, the SiO_2 sacrificial layer under the coupons was removed by wet etching using a dilute hydrofluoric (DHF) acid solution, followed by liquid- CO_2 supercritical point drying, which prevented the coupons from falling and sticking to the bottom Si handle substrate because of the surface tension of the rinse water. Figure 2(b) shows a freestanding TFLN structure. The TFLN coupons were selected and printed. Before selecting the TFLN coupon, a

patterned polydimethylsiloxane (PDMS) elastomer stamp with an optimized curing-agent ratio was prepared. The suspended TFLN coupon was selected by selectively peeling it away from the Si handle substrate using the PDMS stamp at a peeling velocity higher than the critical condition, as shown in Fig. 3(c). Subsequently, the TFLN coupon was printed on the Si target substrate having a 3- μm -thick SiO_2 thermal oxide layer, according to the following process. The selected coupon and Si target substrate were irradiated with O_2 plasma. The TFLN coupon on PDMS was applied to the Si target substrate, and the PDMS was removed at a peel velocity lower than the critical velocity. The TFLN MZI was then printed onto the Si substrate under pressure (Fig. 3(d)). The annealing process was conducted at 250 $^\circ\text{C}$ / 30 min in N_2 gas atmosphere, to enhance the adhesion strength. The printing method was based on hydrophilic O_2 plasma-activated direct bonding, which does not require an adhesion layer such as benzocyclobutene (BCB). Finally, 1.2- μm -thick Au electrodes with an electrode gap of 5 μm and a signal width of 22 μm were formed on the TFLN MZI via maskless exposure lithography, the following electron beam vapor deposition and lift-off process, as shown in Fig. 3(e), and the 1.1-cm-long TFLN MZM was successfully fabricated using the μ -TP process.

3. Experimental results

3.1. Loss measurement

The propagation losses of the micro-transfer-printed TFLN waveguides were measured using the cutback method. Four channels of 1.1-cm-long coupons consisting of two TFLN waveguides were bonded on two Si target substrates. Both sides of the substrates were manually cleaved to obtain 10-mm-long TFLN waveguides. One substrate was then cleaved into three pieces, and each piece contained TFLN waveguides of lengths 3, 3, and 4 mm. The cutback structures printed on the Si substrates are shown in Fig. 4(a). The transverse electric (TE)-polarized continuous wave (CW) light was launched into cutback devices of different lengths through edge coupling from a single-mode lens-tipped fiber using an amplified spontaneous emission light source. The output power was measured using a spectrum analyzer. An example of the transmitted spectrum of a 10-mm-long waveguide is shown in Fig. 4(b). From the measured spectra of the cutback devices with different lengths, the values at a wavelength of 1550 nm were extracted and plotted in Fig. 4(c) as a function of waveguide length. From the linearly fitted curve of the measured transmittance, the propagation loss of μ -TP integrated TFLN waveguides was estimated to be 0.7 dB/cm. The measured data were scattered to some extent because of the variation in edge coupling loss; however, propagation loss was comparable to that reported in [23]. Therefore, the measured propagation loss is primarily attributed to the sidewall roughness, assuming that the micro-transfer printing process causes no significant loss.

3.2. V_π measurement

To measure the half-wave voltage V_π of the μ -TP integrated TFLN MZM, a triangular voltage was applied to the test device through the GSG co-planer electrode at a repetition rate of 100 kHz in a push-pull manner. The amplitude of the waveform was ± 2 V with an offset of -3 V. The offset was applied to compensate for the phase errors between each arm of the TFLN MZI due to the process variation although the arms of the MZI are symmetric. The TE-polarized CW light was coupled to the facet of the modulator using a tunable laser diode (TLD) at a wavelength of 1550 nm. The optical output was detected using a photodetector (PD) and observed using a real-time oscilloscope. Figure 5(a) shows the temporal waveform of the applied voltage and normalized transmittance of the optical signal. From this result, the change of the transmittance as a function of the applied voltage was obtained, as shown in Fig. 5(b). The V_π is estimated to be 2.7 V, which is the lowest half-wave voltage of the micro-transfer printed TFLN modulator reported so far. This is attributed to the long coupon length of 11 mm. For the 9.9-mm-long

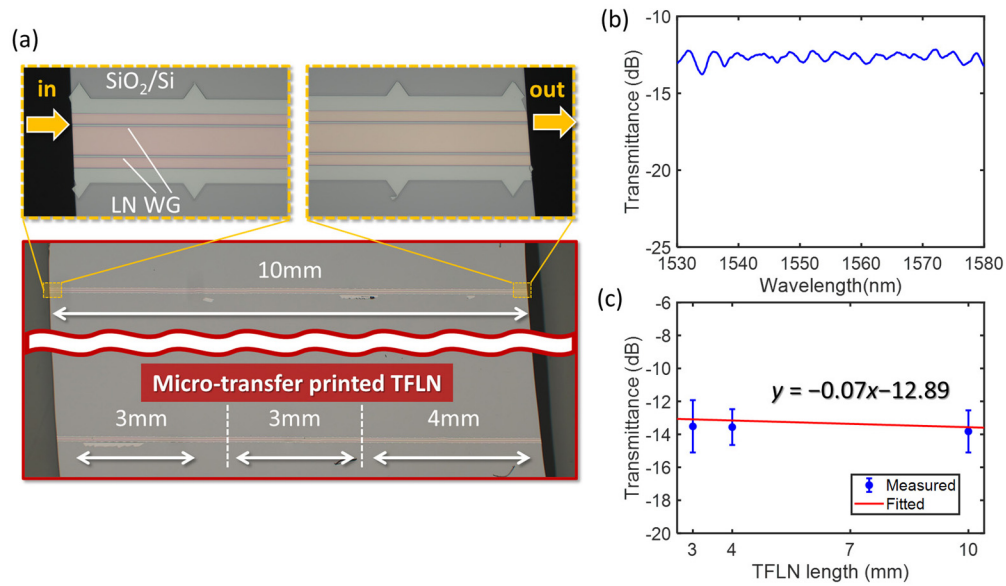


Fig. 4. (a) Micro-transfer printed TFLN waveguides with different device lengths for loss evaluation by cut-back method. (b) Transmitted spectrum of 10-mm-long micro-transfer printed TFLN waveguide. (c) Transmittance of micro-transfer printed TFLN waveguides with different lengths.

phase shifters of the MZI in the push-pull configuration, the half-wave voltage length product $V_{\pi}L$ was 2.7 V-cm, which is slightly higher than the simulated value of 2.5 V-cm in Fig. 5(c). This difference is caused by fabrication error, such as the width and thickness of the slab structure of the waveguide and the gap between electrodes. The insertion loss of the fabricated modulator was 0.97 dB, which included the 0.77 dB propagation loss of the 11-mm-long device and the 0.1 dB simulated excess loss of the MMI coupler.

3.3. High-speed measurement

The small-signal EO bandwidth of the micro-transfer printed TFLN MZMs with different phase-shift lengths of 3.0 and 9.9 mm were characterized using the measurement setup shown in Fig. 6(a). TE-polarized CW light with a wavelength of 1550 nm from a TLD was coupled to the test device through the polarization controller. The output optical power was amplified using an erbium-doped fiber amplifier (EDFA), detected by a PD, and collected using a vector network analyzer (VNA). Figure 6(b) shows the measured EO bandwidth of the micro-transfer-printed TFLN MZM up to 50 GHz, which was normalized to that measured at 1 GHz. The 3-dB EO bandwidth of the 3.0-mm-long device exceeded 50 GHz, limited by the maximum frequency of our VNA. The 3-dB EO bandwidth of the 9.9-mm-long device was approximately 28 GHz, which was limited by the mismatch of the group indices for lightwave and microwave of 2.2 and 2.1, respectively and low characteristic impedance of the GSG electrode of 40 Ω , and a quick EO roll-off was induced by a large microwave absorption of the non-doped bare Si substrate with a low resistivity of 1–20 Ω cm, which can be mitigated by replacing the substrate with a low-k substrate or high-resistivity Si substrate with lower microwave absorption [24]. Undercut structures can further reduce the microwave absorption and be easily realized by using the μ -TP technique [25,26]. In addition, design optimization of the GSG electrode through impedance and velocity matching of the optical and electrical signals can improve the EO performance.

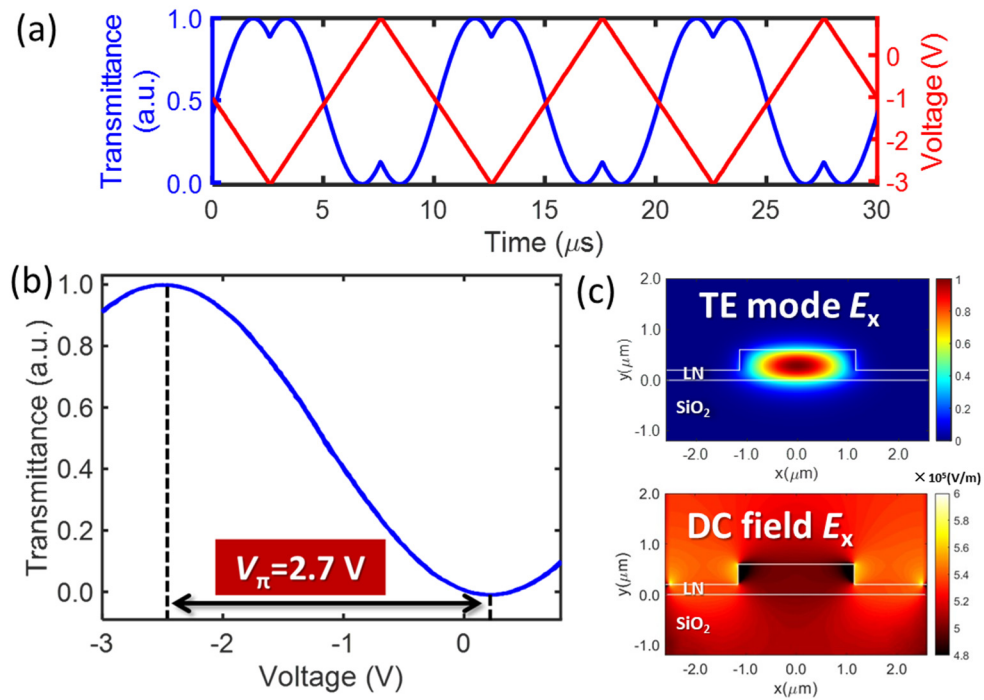


Fig. 5. (a) Temporal evolution of normalized transmittance of light with applied triangular voltages at a repetition rate of 100 kHz. (b) Normalized transmittance as a function of applied voltage. (c) Simulated cross-sectional field distribution of optical mode and DC electric field of TFLN waveguide.

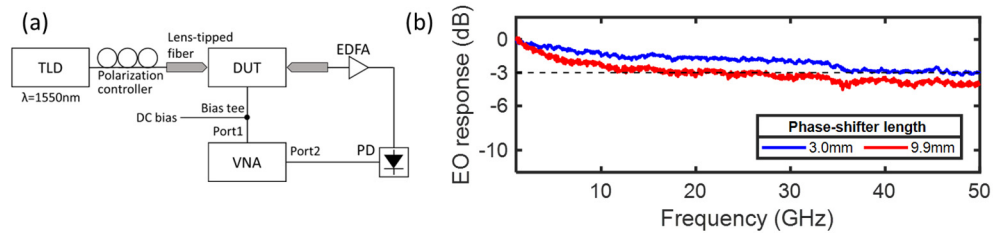


Fig. 6. (a) Measurement setup of EO response. (b) EO response of micro-transfer printed TFLN MZM with different phase-shift lengths of 3.0 and 9.9 mm for bandwidths of up to 50 GHz.

Figure 7(a) shows the measurement setup of the eye diagram of the modulator. A pulse pattern generator (PPG) generated a 50 Gbps NRZ signal using a pseudo-random bit sequence (PRBS7), which was applied to the electrode of the test device using an electrical amplifier. A peak-to-peak voltage of 3.0 V was applied. The electrical signal from the PD was observed using a sampling oscilloscope, and an eye diagram for 50 Gbps was obtained, as shown in Fig. 7(b), which confirms the high-speed operation of the micro-transfer-printed TFLN modulator.

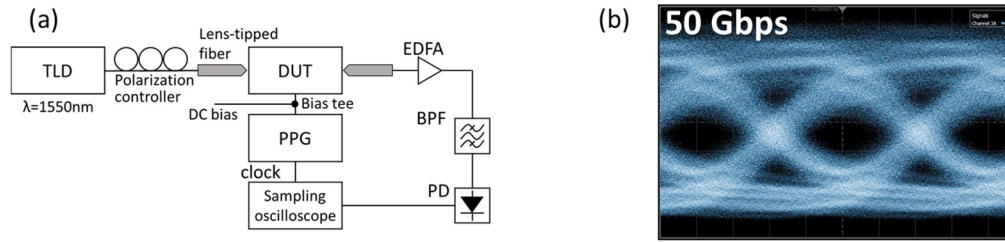


Fig. 7. (a) Measurement setup of eye diagram. (b) Eye diagram for 50 Gbps NRZ signal using PRBS7 measured for micro-transfer printed TFLN MZM with 3.0-mm-long phase shifter.

4. Discussion

Table 1 summarizes the properties and performances of various micro-transfer-printed TFLN modulators. The μ -TP TFLN modulator shown in the first line of Table 1 requires a driving voltage as large as 14.8 V due to the limited 2-mm length of the LN coupons [17] originating from the insufficient mechanical stability of the monolithic LN tether structure. This problem may be solved by using a resist tether with large mechanical stability, thereby reducing the half-wave voltage V_π to 3.3 V by printing the unpatterned 10-mm-long TFLN coupon [21]. Further improvement of V_π may also be difficult due to the weak confinement of optical mode into an unpatterned LN structure. In this work, the size of the TFLN coupon of $11\text{ mm} \times 100\text{ }\mu\text{m}$ was the largest reported so far, and the design optimization of the monolithic LN tether enabled stable suspension and selection. In addition, large bandwidths of $> 28\text{ GHz}$ and V_π as small as 2.7 V with $V_\pi L$ of 2.7 V-cm were achieved simultaneously owing to the strong optical confinement of light in the etched LN waveguide. The $V_\pi L$ can be decreased to 1.9 V-cm by reducing the electrode gap to $4\text{ }\mu\text{m}$ and the driving voltage to less than 2 V, which is achievable. In addition, introducing high-permittivity cladding and a transparent oxide film can further decrease L to less than 1 V-cm [27,28].

Table 1. Comparison of micro-transfer-printed TFLN modulators

Coupon size (Phase-shifter-length)	Tether type	LN waveguide etching	Platform	Modulation type	Insertion loss (dB)	Efficiency V_π or Q	Bandwidth (GHz)	Ref.
2 mm $\times$ 30 μm (2 mm)	LN	No	LN/SiN	MZI	3.3	$V_\pi = 14.8\text{ V}$	> 50	[17]
0.7 mm $\times$ 30 μm (400 μm)	Resist	No	LN/Si	MRR	1.5	$Q = 11000$	16	[18]
100 $\mu\text{m} \times 100\text{ }\mu\text{m}$ (N/A)	Resist	Yes	LN/SiN	MRR	N/A	$Q = 32000$	N/A	[19]
6.5 mm $\times$ 100 μm (5.0 mm)	LN	Yes	LN/Si	MZI	N/A	$V_\pi = 5.0\text{ V}$	N/A	[20]
10 mm $\times$ 30 μm (10 mm)	Resist	No	LN/SiN	MZI	> 4.5	$V_\pi = 3.2\text{ V}$	< 2	[21]
2.4 mm $\times$ 100 μm (2.4 mm)	Resist	No	LN/SiN	MZI	8	$V_\pi = 43.6\text{ V}$	N/A	[29]
11 mm $\times$ 100 μm (9.9 mm)	LN	Yes	LN/Si	MZI	0.97	$V_\pi = 2.7\text{ V}$	28	This work
4 mm $\times$ 100 μm (3.0 mm)	LN	Yes	LN/Si	MZI	0.48	$V_\pi = 8.9\text{ V}^a$	> 50	This work

^a V_π is estimated from the measurement result in Fig. 5(b)

5. Conclusion

In this study, by optimizing the design of monolithic LN tethers and coupon structures, patterned waveguides of TFLN MZM on coupons with up to 1.1-cm length were successfully integrated on the Si target substrate using μ -TP. The fabricated μ -TP integrated TFLN MZM had a low insertion loss of 0.97 dB, which included the propagation loss of waveguides of 0.7 dB/cm measured using cut-back method. To the best of our knowledge, the half-wave voltage V_π of 2.7 V is the lowest reported value of μ -TP TFLN modulator and is attributable to the strong optical confinement of the etched LN waveguide and 9.9-mm-long EO phase shifter length. In addition, the 3.0-mm-long μ -TP TFLN MZM exhibited high-speed operation at bandwidths exceeding 50 GHz, as observed in the eye diagram for 50 Gbps NRZ signal.

By printing the pre-structure-defined TFLN MZI, we have demonstrated the advantage of μ -TP as a heterogeneous integration technology, in which the completed devices could be flexibly printed on any material platforms such as Si, quartz, sapphire or other low-k substrates, and air-suspended structure [25] after material-specific device process. The future challenge is to integrate multiple TFLN coupons as modulators including electrodes, which have wide-capacity-loaded electrode for a large EO bandwidth with low V_π . The proposed micro-transfer-printed TFLN MZM provides high-speed and low-loss modulation function with low driving voltage on any photonic platform, including Si and SiN. The monolithic tether approach to print pre-structure-defined TFLN coupons demonstrated in this work is also applicable to μ -TP of thin-film lithium tantalate [30]. By designing interlayer couplers with simple inverse taper structures underneath the photonic layer, the lightwave is fully transferred to a micro-transfer-printed TFLN MZM with low insertion loss [11]. By using our in-house manufactured μ -TP equipment, the 3σ accuracy of the waveguide-to-waveguide alignment less than 0.5 μ m was reported in [15], and low-loss transition of 0.1 dB with the misalignment-tolerant design between Si and printed TFLN is achievable by using inverse-tapered inter-layer couplers. We believe that the heterogeneous integration of TFLN on SiPh using μ -TP, as in this work, is an indispensable technology toward next-generation 3.2 T optical transceivers.

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Disclosures. The authors declare no conflicts of interest.

Data Availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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